

Computational Physics: Materials Science

Exercise 7

The 2026 sodium-cluster challenge

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The goal of this exercise is to generate the Na_{76} cluster with the lowest potential energy under the Morse potential of exercise sheet 3 (the Girifalco–Weizer parameters for sodium). This is a *global*-optimisation problem on the potential-energy surface (PES) of a 76-atom cluster, which has an exponentially large number of local minima. The strategy follows the four parts of the sheet: (a) build a physical initial structure and melt it with molecular dynamics (MD); (b) relax both the cold and the molten structure with several ASE local optimisers and compare them; (c) assess how close the resulting local minimum is to the global one; and (d) use temperature explicitly — simulated annealing and basin hopping — to search for the lowest-lying minimum. The Atomic Simulation Environment (ASE) drives the dynamics and the optimisers. An AI assistant (Claude) was used to develop the ASE Morse calculator, the optimisation and global-search drivers and the plotting; the choice of methods, their interpretation and this report are my own.

Implementation

Potential

The Morse potential models the cohesive bonding of the sodium cluster through a short-range repulsion and an exponentially decaying attraction,

$$V(r) = D \left[e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)} \right] \quad (r < r_{\text{cut}}), \quad \mathbf{F}_i = - \sum_{j \neq i} \frac{dV}{dr} \frac{\mathbf{r}_{ij}}{r_{ij}}, \quad (1)$$

with $D = 0.063\,34\,\text{eV}$, $r_0 = 5.336\,\text{\AA}$, $\alpha = 0.589\,93\,\text{\AA}^{-1}$ and a hard cutoff at $r_{\text{cut}} = 14.0\,\text{\AA}$ (the pair energy is set to zero beyond r_{cut} ; no shift is applied, matching the literal definition on the sheet). The minimum of a single pair lies at r_0 with depth D , i.e. $V(r_0) = -D$. The dimensionless range parameter is $\rho_0 = \alpha r_0 = 3.148$, a *moderate* range for which the global minima of Morse clusters tend to be compact, polytetrahedral/icosahedral packings rather than open crystalline fragments.

The potential is implemented as a subclass of `ase.calculators.Calculator` that evaluates Eq. (1) for all pairs within the cutoff (a vectorised NumPy kernel, with an optional numba twin for speed; for $N = 76$ the cost is negligible) and writes energy and forces into the `results` dictionary, so the ASE integrators and optimisers use exactly this potential. The analytic forces were verified against a central finite-difference of the energy (maximum deviation $\sim 10^{-10}\,\text{eV/}$), and the single-pair check returns $V(r_0) = -D$ exactly. The cluster is placed at the centre of a $60\,\text{\AA}$ cubic box with periodic boundaries switched off (`pbcs=False`); there is therefore no periodic image at all, and the vacuum requirement of the sheet is satisfied trivially (the final cluster spans $\approx 21.6\,\text{\AA}$, leaving a gap of $\approx 19\,\text{\AA}$ on each side, well beyond r_{cut}).

Initial structures and search methods

Three seed geometries are used: a *face-centred-cubic (fcc) sphere* (the 76 atoms closest to the centre of an fcc lattice with nearest-neighbour spacing r_0 , the most compact ordered seed); a *Mackay icosahedron* (the 76 innermost atoms of a four-shell icosahedron); and a *random compact* sphere. The building blocks of the search are:



- **Robust relaxation.** A two-stage local relaxation FIRE→LBFGS is used everywhere a quick, reliable quench to a minimum is needed: FIRE is robust far from a minimum, LBFGS gives fast quadratic convergence in the final approach.
- **MD heating (a).** A Langevin thermostat ramps the temperature in stages so that the cold seed first relaxes and then melts.
- **Local optimisers (b).** Five ASE optimisers — BFGS, LBFGS, FIRE, MDMin and GoodOldQuasiNewton — are run from both the cold seed and a molten snapshot, recording the energy and the maximum force at every step.
- **Minima ensemble (c).** Twelve independent relaxations from random starts give a distribution of local-minimum energies.
- **Global search (d).** Simulated annealing (melt then slowly quench, several restarts) and basin hopping. The basin-hopping move set combines a Cartesian “shake” with a *worst-atom relocation* move (the most weakly bound atom is moved to a surface gap), and new minima are accepted by a Metropolis criterion.

Simulation parameters

N	D [eV]	r_0 [Å]	α [Å ⁻¹]	r_{cut} [Å]	$\rho_0 = \alpha r_0$	box [Å]	Δt [fs]
76	0.06334	5.336	0.58993	14.0	3.148	60	5

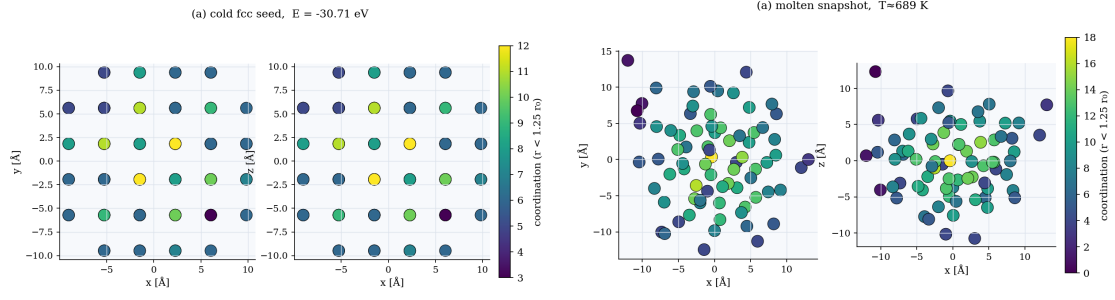
Table 1: Potential and simulation parameters. The MD uses a Langevin thermostat ($\gamma = 0.01 \text{ fs}^{-1}$); local relaxations converge to a maximum force $f_{\text{max}} < 1 \times 10^{-3} \text{ eV/\AA}$ during the search, and the final hand-in structure is tightened to $f_{\text{max}} < 1 \times 10^{-4} \text{ eV/\AA}$. The global search uses 12 random-start minimisations, 6 annealing restarts and 250 basin-hopping steps.

Results

(a) Initial structure and MD melting

The cold fcc seed has an (unrelaxed) energy of -30.71 eV (Fig. 1a); its atoms sit on a regular lattice with a high-coordination core and low-coordination corners. Heating it with the Langevin thermostat (Fig. 2) shows two clearly separated stages. At low temperature the potential energy first *drops* to $\approx -36 \text{ eV}$ as the lattice seed relaxes and contracts into a more compact arrangement. As the temperature is then ramped towards $\approx 700 \text{ K}$, the energy rises back above the cold-seed level and fluctuates around $\approx -28 \text{ eV}$: the cluster has melted. The molten snapshot at $T \approx 689 \text{ K}$ (Fig. 1b) is visibly disordered, occupies a larger volume than the seed, and contains both very high- and very low-coordinated atoms — the structural disorder that lets the system escape the ordered seed basin.





(a) Cold fcc seed, $E = -30.71$ eV.

(b) Molten snapshot, $T \approx 689$ K.

Figure 1: (a) Initial fcc-sphere seed (left) and a snapshot after MD heating (right), both shown in two projections and coloured by coordination number (neighbours within $1.25 r_0$). The ordered seed melts into a disordered, more extended cluster.

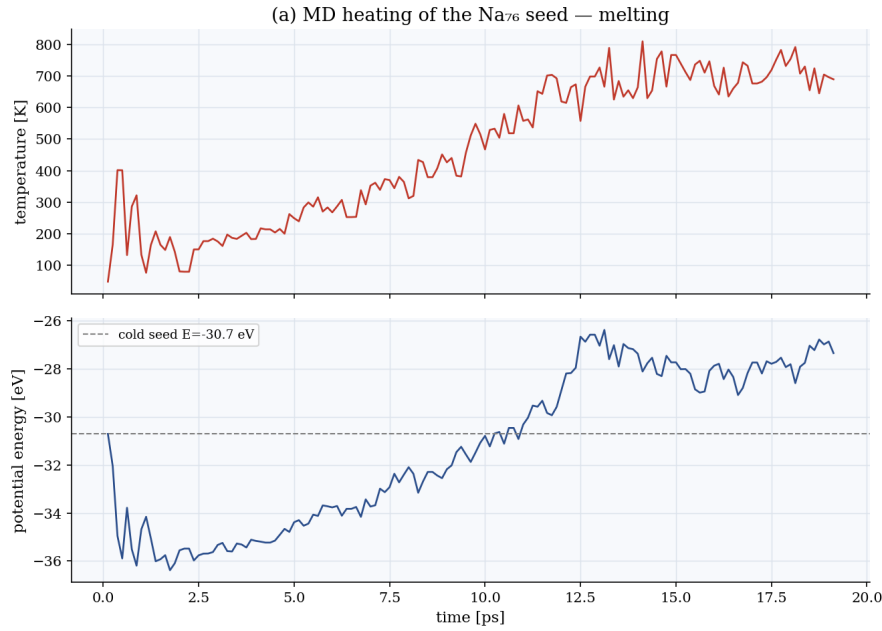


Figure 2: (a) MD heating of the Na_{76} seed. Top: instantaneous temperature ramped from ≈ 50 K to ≈ 700 K. Bottom: potential energy. The energy first drops to ≈ -36 eV as the cold seed relaxes, then rises above the cold-seed reference (dashed, -30.7 eV) as the cluster melts.

(b) Local optimisers and how they differ

Figure 3 compares the five optimisers, started once from the cold seed (left) and once from the molten snapshot (right); the lower panels show the maximum force on a logarithmic axis with the convergence criterion $f_{\max} = 1 \times 10^{-3} \text{ eV/\AA}$ marked.

From the ordered seed. BFGS, FIRE, MDMin and GoodOldQuasiNewton all relax the seed quickly into the same basin at ≈ -37.3 eV, but with markedly different cost: the damped-MD method FIRE reaches the criterion in the fewest steps, followed by MDMin and BFGS, while GoodOldQuasiNewton converges in energy yet stalls in force and does not reach $f_{\max}$ within the step budget. LBFGS is the outlier: started from the ordered, soft-mode seed it becomes *unstable*, its energy oscillating in a band around -31 eV and its force never falling below $\approx 8 \times 10^{-2} \text{ eV/\AA}$. This illustrates the



central difference between the methods: the quasi-Newton optimisers (BFGS, LBFGS, GoodOldQuasiNewton) build an approximate Hessian from successive forces and take near-optimal steps once the PES is locally quadratic, but a poor initial Hessian estimate can send them off course; the damped-MD methods (FIRE, MDMin) follow Newtonian dynamics with friction and are far more robust on rough or near-degenerate starting points.

From the molten snapshot. All five optimisers descend smoothly to a *deeper* basin at ≈ -37.6 eV — about 0.3 eV below the relaxed seed. FIRE again converges in the fewest force evaluations, BFGS next, and MDMin and LBFGS are the slowest to reach $f_{\max}$; GoodOldQuasiNewton is again capped before convergence. The key physical observation is that relaxing a *disordered* structure reaches a lower minimum than relaxing the ordered seed: a single local optimisation only finds the basin it starts in, so starting from many different (and in particular melted) configurations is what uncovers lower minima. This directly motivates parts (c) and (d).

(b) Local optimizers: convergence from ordered vs disordered start

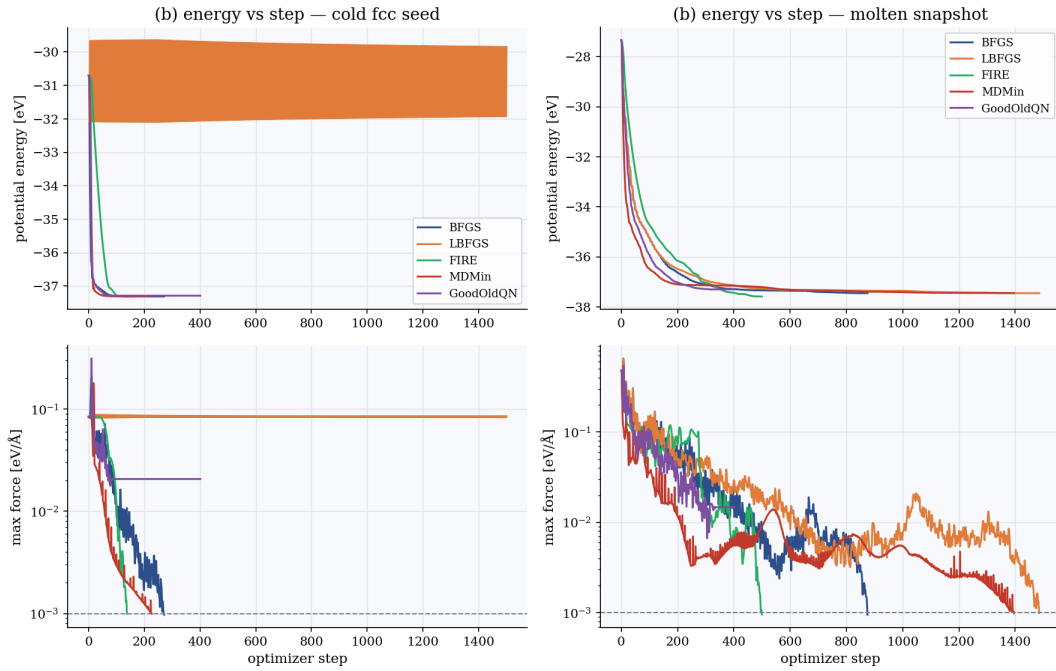


Figure 3: (b) Local optimisers started from the cold fcc seed (left) and from a molten snapshot (right). Top: potential energy vs. optimiser step. Bottom: maximum force (log axis; dashed line = 1×10^{-3} eV/Å). From the ordered seed LBFGS (orange) fails to converge while the others reach ≈ -37.3 eV; from the molten start all reach the deeper ≈ -37.6 eV basin, with FIRE fastest and LBFGS slowest.

(c) Is the local minimum close to the global minimum?

A single relaxation cannot by itself certify a global minimum. To gauge how close we are, twelve independent relaxations were run from random starting structures and their final energies histogrammed (Fig. 4). Most random starts fall into a band of shallow minima between ≈ -37.47 eV and ≈ -37.32 eV, but *one* independent random start reaches -37.660 eV — the same low-lying basin later confirmed by the global search in



part (d). The fact that an unbiased random start, with no knowledge of the answer, independently lands in this basin is the main reason to believe it is at or very near the global minimum: the lowest basin is reachable from the “raw” PES and sits well below the cloud of competing minima.

Confidence can be increased further by (i) running many more independent searches and checking that none goes lower, (ii) confirming that *different* methods (random quench, annealing, basin hopping) all converge to the *same* energy and structure, and (iii) comparing against tabulated global minima of Morse clusters at this range parameter. In this work the independent random hit and the basin-hopping result agree to within a few meV, which is the practical evidence used here. A rigorous proof of globality is not possible by search alone.

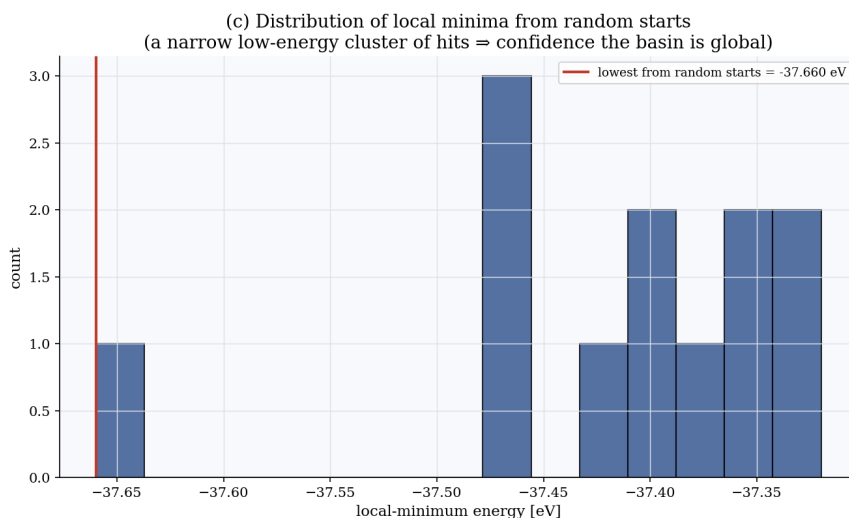


Figure 4: (c) Distribution of local-minimum energies from twelve independent random-start relaxations. Most starts give shallow minima ($\approx -37.47 \dots -37.32$ eV); a single independent start reaches the low-lying basin at -37.660 eV (red line), the same minimum found by the global search.

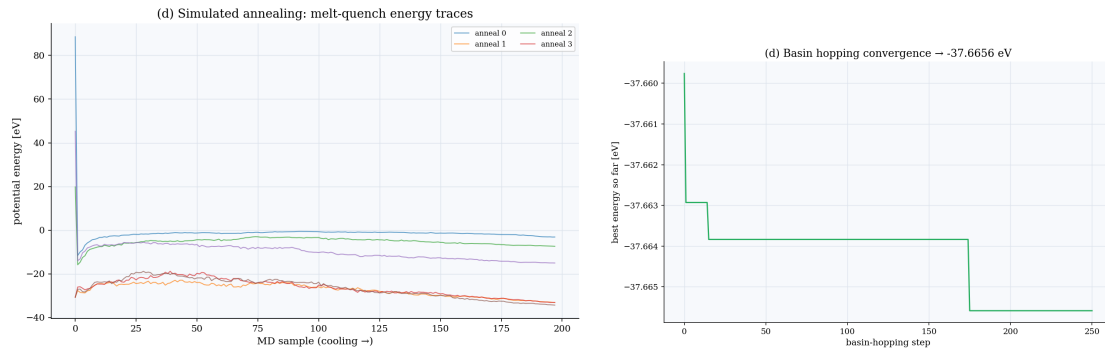
(d) Using temperature: annealing and basin hopping

The strategy that uses temperature explicitly is *simulated annealing*: heat the cluster until it is liquid so it can rearrange over barriers, then cool it slowly so it settles into a low-energy basin. Figure 5a shows the potential-energy traces of six melt-quench restarts; the curves start from hot, high-energy configurations and descend as the temperature is lowered, with different restarts ending in different basins (the lowest traces reach ≈ -34 eV during cooling before the final relaxation). The spread confirms the lesson of part (c): a single anneal does not reliably reach the global minimum, which is why several restarts and an additional basin-hopping stage are used. **In a real experiment this is literally thermal annealing** — a sample is heated to let its atoms diffuse and reorganise, then cooled slowly so it relaxes towards its ground-state structure instead of freezing into a defective one.

Basin hopping refines the search by repeatedly perturbing the current structure (a Cartesian shake or a worst-atom relocation), relaxing to the nearest minimum, and accepting or rejecting by a Metropolis rule. Figure 5b shows the best energy found so far decreasing



in distinct steps to -37.6656 eV over 250 hops; the relocation move (moving the most weakly bound atom to a surface gap) is what produces the largest improvements, because it lets the search cross between structural motifs that a small Cartesian shake cannot reach.



(a) Simulated-annealing energy traces.

(b) Basin-hopping convergence.

Figure 5: (d) Temperature-driven global search. (a) Six melt-quench restarts cool from hot disordered states into a spread of basins. (b) Basin hopping lowers the best-so-far energy in steps to -37.6656 eV.

The winning cluster

The lowest configuration located by the search is shown in Fig. 6. After a final tight relaxation ($f_{\max} < 1 \times 10^{-4}$ eV/Å) the handed-in structure has

$$E = -37.6636 \text{ eV} = -0.4956 \text{ eV/atom} = -7.82 D, \quad (2)$$

i.e. a cohesive energy of 7.82 pair-well depths per atom. (The basin-hopping search reported -37.6656 eV in Fig. 5b; the ≈ 2 meV difference to the handed-in value is at the level of the force-convergence tolerance.) The cluster is compact and roughly spherical, with a high-coordination core (up to ≈ 16 neighbours within $1.25 r_0$) and a low-coordination surface (down to ≈ 6), as expected for a moderate-range Morse cluster. It is saved as `Na76_cluster_StefanusMichael.traj`.



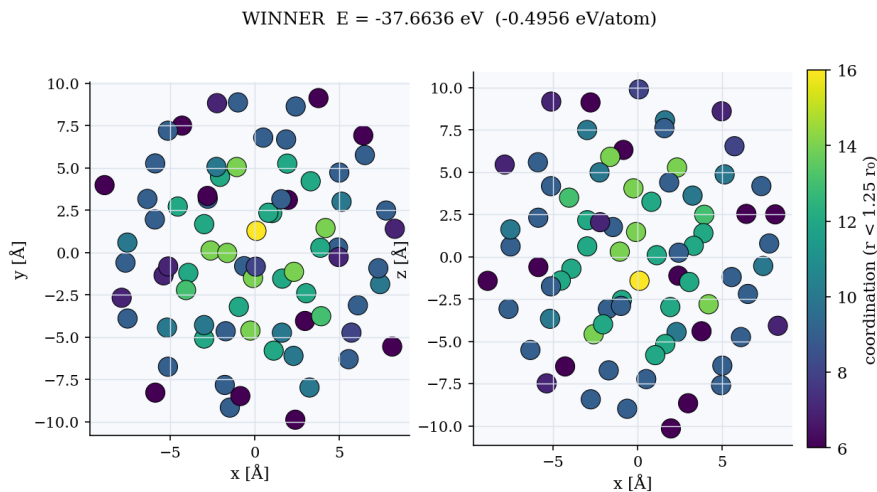


Figure 6: Lowest-energy Na_{76} cluster, $E = -37.6636 \text{ eV}$ (-0.4956 eV/atom), shown in two projections and coloured by coordination number. The dense, high-coordination core is surrounded by a lower-coordination surface.

Key takeaways

- **Locating local minima.** Two families were used. *Quasi-Newton* methods (BFGS, LBFGS, GoodOldQuasiNewton) approximate the Hessian from successive forces and converge fast near a quadratic minimum, but are sensitive to the starting point (LBFGS failed on the ordered seed). *Damped-MD* methods (FIRE, MDMin) integrate Newton's equations with friction; they are slower in the quadratic region but much more robust on disordered starts — FIRE was the most reliable and fastest overall here.
- **Searching for the global minimum.** A single relaxation only finds the nearest basin. The harder global problem is tackled by sampling many starts (c) and by hopping between basins with temperature (d): simulated annealing (melt then quench slowly) and basin hopping with a relocation move. Confidence that the result is global comes from independent methods agreeing on the same energy and structure, not from a single run.
- **Applications of geometry optimisation.** Beyond this challenge, relaxing a structure yields equilibrium geometries and bond lengths, cohesive and defect-formation energies, and — via the Hessian at the minimum — vibrational frequencies and elastic/thermodynamic properties; transition-state searches between minima give reaction barriers. The optimised structures are also the physical starting points for further MD or property calculations.

Summary

- **Result.** The lowest Na_{76} Morse cluster found has $E = -37.6636 \text{ eV}$ (-0.4956 eV/atom , $-7.82 D$), a compact roughly spherical structure with a high-coordination core.
- **Potential.** A custom truncated Morse ASE calculator ($\rho_0 = 3.148$), verified against finite-difference forces and $V(r_0) = -D$; the cluster sits in vacuum (`pbc=False`, $\approx 19 \text{ Å}$ gap $> r_{\text{cut}}$).



- **Melting (a).** The fcc seed (-30.71 eV) first relaxes, then melts above ≈ 700 K into a disordered cluster.
- **Optimisers (b).** From the cold seed all methods reach ≈ -37.3 eV except LBFGS, which diverges; from the molten start all reach the deeper ≈ -37.6 eV basin, FIRE fastest and LBFGS slowest. Relaxing disordered structures finds lower minima than relaxing the ordered seed.
- **Global search (c,d).** One of twelve random starts independently reaches the -37.660 eV basin; simulated annealing and basin hopping (with a worst-atom relocation move) confirm and refine it to -37.6656 eV. Annealing is the simulation analogue of thermal annealing in a real experiment.

